REVIEW

Collagen peptide supplementation for pain and function: is it effective?

Shiloah A. Kviatkovsky[a,b,c], *Robert C. Hickner*[a,b,d] *and Michael J. Ormsbee*[a,b,d]

Purpose of review
Connective tissue injuries are prevalent in active and aging populations, leading to chronic pain and decreased function. Turnover of this tissue is not well understood, especially as it relates to aging and injury. Supplementation of collagen peptides has been shown to improve connective tissue recovery and pain through increased collagen production.

Recent findings
Collagen peptide supplementation improves pain and function, and upregulates metabolic pathways associated with muscle and tendon growth. Literature from the past 12–18 months supports that these pathways are also involved with increased synthesis and degradation of collagen and other elements of the extracellular matrix. Improvements in body composition and strength have been noted with collagen peptide supplementation when paired with resistance training. Collagen peptide supplements are hydrolyzed into small peptides, termed bioactive peptides, and individual amino acids. These bioactive peptides are associated with the benefits observed with collagen peptide supplementation and may play a critical role in the collagen turnover.

Summary
Collagen peptide supplementation has been shown to promote recovery, decrease pain, and improve strength and body composition when paired with resistance training. These benefits may be attributed to bioactive peptides in collagen peptide supplements. Additional research is warranted to examine the specific effects of these bioactive peptides.

Keywords
bioactive peptides, connective tissue, gelatin, hydrolyzed, sport nutrition

INTRODUCTION

Collagen peptide supplementation has gained popularity in recent years because of the purported benefits on a variety of physiological functions ranging from skin and joint health to recovery from injury and mental wellbeing/cognition [1–4]. Collagen is the most abundant protein in the human body and a major constituent of the extracellular matrix (ECM) in tissue such as skin, tendons, ligaments, and bone [5,6]. Approximately one-third (1/3) of total body protein is collagen, of which 90% is type I collagen [5]. Collagen peptide supplements are made of hydrolyzed collagen/gelatin that have been degraded into individual amino acids and smaller peptides with a lower molecular weight compared to native collagen (3–6 versus 285–300 kDa, respectively). The lower molecular weight allows for transport of small intact peptides (bioactive peptides) into circulation, which are thought to boost endogenous collagen synthesis [1,7]. Collagen peptide supplements have been demonstrated to enhance collagen synthesis by fibroblasts in the ECM, thereby repairing photodamage in skin and reducing joint pain and inflammation in individuals suffering from osteoarthritis and connective tissue injuries [8].

[a]Department of Nutrition and Integrative Physiology, [b]Institute of Sports Sciences and Medicine, Florida State University, Tallahassee, Florida, [c]Center for Aging and Longevity, Geriatrics, University of Arkansas for Medical Sciences, Little Rock, Arkansas, USA and [d]Department of Biokinetics, Exercise and Leisure Sciences, University of KwaZulu-Natal, Durban, South Africa

Correspondence to Michael J. Ormsbee, PhD, Institute of Sports Sciences and Medicine, Florida State University, 1104 Spirit Way, Tallahassee, FL 32306, USA. Tel: +1 850 645 1016; e-mail: mormsbee@fsu.edu

Curr Opin Clin Nutr Metab Care 2022, 25:401–406
DOI:10.1097/MCO.0000000000000870

1363-1950 Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.

KEY POINTS

- Connective tissue injuries are prevalent and difficult to repair.
- Collagen peptide supplementation appears to improve tissue repair, increase physical function, and decrease pain, yet the mechanisms of action are unclear.
- Hydrolyzed collagen peptides contain high concentrations of small bioactive peptides, which are present in blood after consumption and are thought to promote collagen turnover in connective tissue.
- Future research directions should consider collagen peptide supplementation formulations and compare effects matched by bioactive peptide composition.

Tendons and ligaments are the connective tissues that support the musculoskeletal systems and allow for locomotion through force transmission between muscles and bones, and between bones and bones [9]. These connective tissues withstand the greatest tensile and mechanical forces exerted on the human body, placing them at a high risk for degradation and injury. For example, of the estimated 33 million musculoskeletal injuries reported in United States annually, approximately 50% are to tendons and ligaments [9]. Due to low vascularity and cellularity, these types of injuries are the hardest to repair and take the longest to heal [9]. Furthermore, low blood supply and nutrient delivery inhibit full recovery from musculoskeletal injuries leaving the integrity of the tissue compromised indefinitely (only recovering to 40–70% of the pre-injury strength) [9,10]. Consequently, more than 19% of the United States' adult population experiences chronic pain. Additionally, even without injury, loss of body collagen turnover is thought to begin in the second decade of life and rapidly declines throughout the lifespan [9,10]. Therefore, these types of injuries and the associated pain are a major cause of disability in active as well as aging populations.

COLLAGEN TURNOVER

Collagen tissue turnover is understudied, especially in comparison to other tissue types that degrade with age, such as muscle and bone. This is largely because of the difficulty in measuring collagen turnover. The mechanisms that regulate collagen remodeling (i.e. synthesis and degradation) can vary depending on collagen fiber and tissue type as well as health status. It is relatively well established that fibrillar collagen fiber types are synthesized in fibroblasts via the transcription growth factor beta (TGF-β)/SMAD pathway [11]. Furthermore, TGF-β is known to be a key regulatory pathway after injury, involved in the initial inflammatory response, collagen synthesis, and scar tissue/fibrotic tissue development [11]. Interestingly, although physical activity introduces higher injury risks, mechanical loading has shown to stimulate type I collagen synthesis through activation of the TGF-β/SMAD pathway [11,12]. These forces are also sensed by other elements of the ECM such as matrix metalloproteinases (MMPs), which are enzymes responsible for degrading injured collagen fibers for new network formation [11,12]. In the absence of these forces (i.e. being sedentary), degradation of old and damaged collagen fibers decreases while synthesis remains constant, forming stiff fibrotic tissue rather than new healthy collagen networks [12,13]. Stiff connective tissue is characterized by decreased contractility and diminished force production and strength [12,14]. Loss of strength is directly correlated to increased risk of injury in aging populations [9,10]. A more detailed review of musculoskeletal connective tissue remodeling was recently published by Holwerda and van Loon [15▪].

In addition to force and mechanical loading, aging also plays a role as a major regulator of collagen tissue turnover. It is well established that collagen synthesis declines with age [9,10]. Recently though, Abbot *et al.* [14] used deuterium (D_2O) labeling to demonstrate collagen degradation, rather than synthesis, declines in aged compared with adult mouse connective tissue. Abbot *et al.* [14] found the lack of clearance of old and damaged collagen fibers prior to new network formation led to the formation of stiff fibrotic connective tissue. These novel findings suggest that degradation, rather than synthesis hinders collagen tissue turnover [14]. Hence, the reduction in crosslinking of collagen fiber networks may determine its integrity and strength, with fibrotic tissue development leading to loss of function. These recent findings provided new insight that the health and functionality of collagen fibers in connective tissue are dependent on an optimal balance between degradation and synthesis, not just synthesis, which are heavily influenced by activity level and age [12].

Interestingly, supplementation with collagen peptides has also shown to not only upregulate collagen production but to also regulate MMP activity. For example, dermal fibroblasts supplemented with collagen peptides *in vitro* demonstrated upregulation of the TGF-β/SMAD pathway and procollagen synthesis, while also suppressing MMP-1 and MMP-3 expression [15▪]. It is important to note, while the physiological response of MMP activity

Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.

can vary, MMP-1 and MMP-3 are enzymes involved in unhealthy collagen degradation [16]. Although no human studies have directly measured TGF-β/SMAD and MMP activity in conjunction with collagen peptide supplementation, a landmark study by Shaw *et al.* measured increased collagen production after supplementation via plasma amino-terminal propeptide of type I procollagen (P1NP). Shaw *et al.* [17] measured dose-dependent elevations in plasma P1NP using a revolutionary study design, in which engineered ligaments were treated with the serum of participants who supplemented before and 1 h after consumption of either 5 g/day collagen peptides, 15 g/day collagen peptides, or placebo. Together, these results support that collagen peptide supplementation can theoretically lead to improved collagen tissue turnover via pathways involved in synthesis and degradation, independent of exercise. Naturally, this topic warrants further investigation, but translational studies are challenging in humans.

STRENGTH AND BODY COMPOSITION

There is significant debate as to the effect of collagen peptide supplementation on skeletal muscle hypertrophy. This debate stems from the fact that collagen is not a complete protein (e.g. lacks tryptophan) and, therefore, collagen peptide supplements lack sufficient essential amino acids, especially leucine, necessary to stimulate muscle protein synthesis. Despite this, strength and body composition improvements have been observed in several studies examining the effects of resistance training in combination with collagen peptide supplementation [18,19]. For example, Zdzieblik *et al.* [18] measured significant increases in fat-free mass and decreased fat mass in middle-aged untrained men when supplementing with 15 g/day compared with placebo when combined with a resistance training program over 12 weeks. Although, strength outcomes improved similarly in both the collagen peptides and placebo groups [17]. Similarly, Jendricke *et al.* observed significant improvements in fat-free mass and fat mass in premenopausal women after supplementing with 15 g/day collagen peptides combined with 12 weeks of resistance training compared with placebo [18]. However, contrary to findings from Zdzieblik *et al.* [18], Jendricke *et al.* did report significant improvements in hand grip strength in the collagen peptides groups compared with placebo, even though leg strength improvements were equivalent between groups [17].

To address whether collagen peptide supplements are as effective as protein with a complete amino acid profile, Jacinto *et al.* [20] compared whey protein to a leucine-matched collagen peptide supplement combined with resistance training over 10 weeks in young untrained adults. Although greater increases in muscle thickness were observed in the whey protein group, strength and power outcomes were equivalent between groups [20]. The strength gains with collagen peptide supplementation reported by Jacinto *et al.* are in line with outcomes from muscle proteome analyses in young male individuals who supplemented with either 15 g/day collagen peptides or placebo combined with a resistance training over 12 weeks, reported by Oertzen-Hagemann *et al.* [21]. Although the supplement and treatment groups did not differ significantly in strength measures, significant upregulation of 221 proteins associated with protein metabolism of contractile fibers in the collagen peptide supplement group compared with only 44 in the placebo group were found [21]. Additionally, akin to findings reported by Zdzieblik *et al.* and Jendricke *et al.*, those who supplemented with collagen peptides had significantly greater increases in lean body mass compared with the placebo group [17,18,20]. The increased gene expression of these contractile fibers as well as increased phosphorylation of the PI3K-Akt and MAPK pathways may explain the increases in lean body mass and strength reported in other studies. Furthermore, upregulation of contractile fibers with collagen peptide supplementation may also explain how strength gains without increases in muscle size have been noted when supplementing with collagen peptides compared with whey protein [20,21].

Improvements in tendon strength and cross-sectional area have been observed in addition to muscular strength gains. For example, Jerger *et al.* [22] recently examined the effects of 5 g/day collagen peptide supplementation combined with resistance training over 14 weeks on Achilles tendon size and muscular strength in young healthy male individuals. They demonstrated that collagen peptide supplementation yielded significantly greater tendon and plantar flexor cross-sectional area compared with controls. However, tendon stiffness and muscular strength improved similarly in both groups with resistance training [22]. These increases in tendinous tissue synthesis when combined with a resistance or rehabilitative program may also enhance recovery from injury via improved collagen production. It is also important to note that the literature does not support muscle hypertrophy gains, despite strength gains with collagen peptide supplementation. The strength gains observed are most likely because of increased contractile and collagen fiber synthesis as mentioned previously, rather than increases in sarcomere or myofibrillar number.

1363-1950 Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.

PAIN AND FUNCTION

Several studies have observed improvements in pain and function post injury with collagen peptide supplementation, but few studies have evaluated the effects of collagen peptide supplementation on the presence of pain in the absence of a diagnosed injury. Our lab recently investigated this by examining the effects of 6 and 9 months of 10 or 20 g/day of collagen peptide supplementation compared with placebo on chronic pain, function, and mental health-related quality of life in older recreationally active adults. Surprisingly, we found significant improvements in the Knee Injury & Osteoarthritis Outcome Score (KOOS) measure of physical function in the 10 g/day collagen peptide group and not in the 20 g/day or placebo groups after 6 months of supplementation. Additionally, higher frequency exercisers in the 10 g/day collagen peptide group had significant improvements in pain compared with 20 g/day and placebo [3]. Furthermore, our study was the first to utilize the Veterans Rand 12 Item Health Survey (VR-12) in conjunction with collagen peptide supplementation and found significant improvements in mental component scores (MCS) with 10 g/day after 3 months of supplementation, which carried over until the 9-month study end point [4]. And lastly, physical component scores (PCS) were significantly improved with 20 g/day collagen peptide supplementation after 9 months in female individuals but not male individuals, indicating potential sex differences in the effects of collagen peptide supplementation. Although this is a relatively unexplored topic, sex hormones have shown to influence collagen metabolism, and women have shown to have a lower collagen synthetic rate compared with age-matched men [23]. For this reason, perhaps women need to supplement with a larger dose of collagen peptides for longer durations than men. Hence, the topic of sex differences in collagen peptide supplementation warrants further exploration.

Alternatively, with respect to dosage, Bongers *et al.* did not detect significant differences between treatment and control groups in knee pain, function, inflammation, or biomarkers of bone synthesis with 10 g/day collagen peptide supplementation over 12 weeks in older adults with self-reported knee pain [24]. As improved function and pain have only been observed with acute collagen peptide supplementation in the presence of a specific injury, their lack of significant findings may be because of the short study duration, especially in the absence of a specific injury [24,25]. Also, prior studies report improvements in pain and function when combined with rehabilitative or resistance training programs [26,27]. Due to the poor vascularity of connective tissue and because collagen synthesis is stimulated by force, physical activity seems to be an important modulator for the therapeutic effects of collagen peptide supplementation. In support of this hypothesis, Prowting *et al.* observed faster recovery 24 h post muscle damaging exercise with consumption of 15 g/day of collagen peptide supplements consumed over 7 days prior, compared with control in resistance-trained men [24].

BIOACTIVE PEPTIDES

Although the body of literature on the benefits of supplementing with collagen peptides is expanding, information on the mechanism of action of collagen peptide supplementation is lacking. An emerging hypothesis is that specific dipeptides and tripeptides, especially those containing hydroxyproline (bioactive peptides), upregulate gene expression for the synthesis of type I collagen [28,29]. Collagen, especially type I fibers (in tendons and ligaments), is made of triple helical structures composed of Gly-X-Y (glycine-X-Y), with glycine (Gly) constituting 1/3 of the helix, and proline (Pro) and 4-hydroxyproline (Hyp) frequently occupying the X and Y positions, respectively (Gly-Pro-Hyp) [5]. Other amino acids can combine with Gly at the X and Y positions such as arginine, serine, and hydroxylysine as well [5]. These amino acids alone and in combinations of small peptides were first shown to pass through the gastrointestinal tract intact in a seminal study by Iwai *et al.*, with peak concentrations in blood at 60 min post ingestion [30]. Although these findings have been replicated in numerous studies since, measuring intact peptides in human blood can be challenging because of the low molecular weight of these bioactive peptides. Additionally, the presence and concentration of these peptides is highly variable between products, depending on source, length of supplementation, mode of hydrolysis, and other proprietary factors, which could explain the variability in results across studies. Therefore, Larder *et al.* more recently utilized digests from the gut, rather than blood, to compare post consumption bioactive peptide concentrations between two different collagen peptides products [31]. The digests of both collagen peptide formulations contained significantly elevated concentrations of Ala-Hyp, Pro-Hyp, Gly-Pro-Hyp, and Pro-Hyp-Gly, although other bioactive peptide concentrations varied between collagen peptides products [31]. Their work validated the measurements previously found in blood against digestive juices and signified the importance of considering the formulation of collagen peptide products when evaluating their potential health benefits.

Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.

Collagen-derived bioactive peptides, specifically those containing Hyp (e.g. Ala-Hyp, Pro-Hyp-Gly, and Gly-Pro-Hyp) are resistant to enzymatic cleavage by peptidases during digestion and are, therefore, transported into circulation intact [28,31]. These bioactive peptides are thought to act as second messengers regulating mechanisms for ECM degradation, synthesis, inflammation, and repair [28,31]. For example, Ide *et al.* recently analyzed mouse tendon cell activity when treated with various Hyp-containing bioactive peptides and found significant upregulation of ERK phosphorylation, increased tenocyte cell proliferation, and increased collagen type I and fibronectin mRNA levels with Pro-Hyp, compared with Ala-Hyp-Gly, Leu-Hyp-Gly, and Hyp-Gly [28]. Additionally, they demonstrated Pro-Hyp transport into tenocytes to be superior compared with the aforementioned peptides and occurs via endocytosis and translocation pathways rather than strictly through PEPT1 transport into adult tenocytes, as previously thought [28]. Similarly, Nogimura *et al.* measured elevated plasma concentrations of Pro-Hyp compared with Hyp-Gly in a rat model evaluating the effects of collagen peptide supplementation postingestion compared with control [32]. Interestingly, they also measured these peptide concentrations in cerebral spinal fluid and found Pro-Hyp but no Hyp-Gly. This indicates that Pro-Hyp can pass through the blood–brain barrier whereas Hyp-Gly cannot. The elevated concentrations of Pro-Hyp were also associated with increased dopaminergic activity and decreased depressive symptoms, which appears to support our findings of improved VR-12 MCS scores when supplementing with collagen peptides [4,32]. These effects were not present with ingestion of the amino acid components of Pro-Hyp, further strengthening the hypothesis that bioactive peptides, rather than the amino acids, are the drivers for the observed therapeutic effects of collagen peptide supplements.

CONCLUSION

Literature over the past 12–18 months supports the contention that supplementation with collagen peptides can improve recovery from injury and pain associated with connective tissue injuries. Recent findings have demonstrated that increased collagen synthetic pathways are upregulated with collagen peptide supplementation, leading to improvements in body composition and strength when combined with exercise or rehabilitative programs. These benefits are associated with increased concentrations of Hyp-containing bioactive peptides in blood after supplementation, although the concentrations of bioactive peptides are source-specific and formulation-specific. Emerging evidence suggests collagen peptide supplementation may improve mental/cognitive health outcomes, with responses to collagen peptide supplementation potentially differing by sex.

Acknowledgements

None.

Financial support and sponsorships

Authors have received grant funding from PB Leiner, part of Tessenderlo group, to study the role of collagen peptide supplementation on pain, function, and connective tissue turnover. PB Leiner manufactures gelatins and collagen peptides, and is the manufacturer of SOLUGEL, the collagen peptide supplement that was used in studies completed in our lab.

Conflicts of interest

There are no conflicts of interest.

REFERENCES AND RECOMMENDED READING

Papers of particular interest, published within the annual period of review, have been highlighted as:
■ of special interest
■■ of outstanding interest

1. Cao C, Xiao Z, Tong H, *et al.* Oral intake of chicken bone collagen peptides anti-skin aging in mice by regulating collagen degradation and synthesis, inhibiting inflammation and activating lysosomes. Nutrients 2022; 14:1622.
2. Koizumi S, Inoue N, Sugihara F, Igase M. Effects of collagen hydrolysates on human brain structure and cognitive function: a pilot clinical study. Nutrients 2020; 12:50.
3. Kviatkovsky SA, Hickner RC, Gipson SD, et al. Collagen peptide supplementation improves measures of activities of daily living and pain in active adults. Proceedings of the Eighteenth International Society of Sports Nutrition (ISSN) Conference and Expo. 2021.
4. Kviatkovsky SA, Hickner RC, Gipson SD, et al. Collagen peptide supplementation improves measures mental component scores of the VR-12 in active adults. Proceedings of the Nineteenth International Society of Sports Nutrition (ISSN) Conference and Expo 2022. In press.
5. Ricard-Blum S. The collagen family. Cold Spring Harbor Perspect Biol 2011; 3:1–19.
6. Lin P, Hua N, Hsu Y-C, *et al.* Oral collagen drink for antiaging: Antioxidation, facilitation of the increase of collagen synthesis, and improvement of protein folding and DNA repair in human skin fibroblasts. Oxid Med Cell Longev 2020; 8031795. doi: 10.1155/2020/8031795.
7. Mobasheri A, Mahmoudian A, Kalvaityte U, *et al.* A white paper on collagen hydrolyzates and ultrahydrolyzates: potential supplements to support joint health in osteoarthritis? Curr Rheumatol Rep 2021; 23:78.
8. Clark KL, Sebastianelli W, Flechsenhar KR, *et al.* 24-Week study on the use of collagen hydrolysate as a dietary supplement in athletes with activity-related joint pain. Curr Med Res Opin 2008; 24:1485–1496.
9. Titan AL, Foster DS, Chang J, Longaker MT. Flexor tendon: development, healing, adhesion formation, and contributing growth factors. Plast Reconstr Surg 2019; 144:639e–647e.
10. Clark ST, Zhu M, Gamble GD, *et al.* Epidemiology of tendon and ligament injuries in Aotearoa/New Zealand between 2010 and 2016. Injury Epidemiol 2020; 7:5.
11. Fleischhacker V, Klatte-Schulz F, Minkwitz S, *et al.* In vivo and in vitro mechanical loading of mouse Achilles tendons and tenocytes—a pilot study. Int J Mol Sci 2020; 21:1313.
12. Mousavizadeh R, Hojabrpour P, Eltit F, *et al.* β1 integrin, ILK and mTOR regulate collagen synthesis in mechanically loaded tendon cells. Sci Rep 2020; 10:1–12.
13. Geervliet E, Moreno S, Baiamonte L, *et al.* Matrix metalloproteinase-1 decorated polymersomes, a surface-active extracellular matrix therapeutic, potentiates collagen degradation and attenuates early liver fibrosis. J Control Release 2021; 332:594–607.

Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.

14. Abbott CB, Lawrence MM, Kobak KA, *et al.* A novel stable isotope approach demonstrates surprising degree of age-related decline in skeletal muscle collagen proteostasis. Function 2021; 2:zqab028.
15. ■ Holwerda AM, van Loon LJC. The impact of CP ingestion on musculoskeletal connective tissue remodeling: A narrative review. Nutrition Reviews 2022; 80:1497–1514.

An up-to-date comprehensive review on collagen peptide supplementation and muscle and connective tissue turnover.

16. Liu Z, Li Y, Song H, *et al.* Collagen peptides promote photoaging skin cell repair by activating TGF-beta/Smad pathway and depressing collagen degradation. Food Nutr 2019; 10:6121–6134.
17. Shaw G, Lee-Barthel A, Ross MLR, Wang B, *et al.* Vitamin C-enriched gelatin supplementation before intermittent activity augments collagen synthesis. Am J Clin Nutr 2017; 105:136–143.
18. Zdzieblik D, Jendricke P, Oesser S, *et al.* The influence of specific bioactive collagen peptides on body composition and muscle strength in middle-aged, untrained men: a randomized controlled trial. Int J Environ Res Public Health 2021; 18:4837.
19. Jendricke P, Centner C, Zdzieblik D, *et al.* Specific collagen peptides in combination with resistance training improve body composition and regional muscle strength in premenopausal women: a randomized controlled trial. Nutrients 2019; 11:892.
20. Jacinto JL, Nunes JP, Gorissen, *et al.* Whey protein supplementation is superior to leucine-matched collagen peptdides to increase muscle thickness during a 10-week resistance training program in untrained young adults. Int J Sport Nutr Exercise Metab 2022; 32:133–143.
21. Oertzen-Hagemann V, Kirmse M, Eggers B, *et al.* Effects of 12 weeks of hypertrophy resistance exercise training combined with collagen peptide supplementation on the skeletal muscle proteome in recreationally active men. Nutrients 2019; 11:1072.
22. Jerger S, Centner C, Lauber B, *et al.* Effects of specific collagen peptide supplementation combined with resistance training on Achilles tendon properties. Scand J Med Sci Sports 2022; 00:1–11.
23. Bennett K, Vincent T, Sakthi-Velavan. The patellar ligament: a comprehensive review. Clin Anat 2021; 35:52–64.
24. Bongers CCWG, ten Haaf DSM, Catoire M, *et al.* Effectiveness of collagen supplementation on pain scores in healthy individuals with self-reported knee pain: a randomized controlled trial. Appl Physiol Nutr Metab 2020; 45:793–800.
25. Prowting JL, Bemben D, Black CD, *et al.* Effects of collagen peptides on recovery following eccentric exercise in resistance-trained males—a pilot study. Int J Sport Nutr Exercise Metab 2020; 31:32–39.
26. Praet SFE, Purdam CR, Welvaert M, *et al.* Oral supplementation of specific collagen peptides combined with calf-strengthening exercises enhances function and reduces pain in Achilles tendinopathy patients. Nutrients 2019; 11:1–16.
27. Dressler P, Gehring D, Zdzieblik D, *et al.* Improvement of functional ankle properties following supplementation with specific collagen peptides in athletes with chronic ankle instability. J Sports Sci Med 2018; 17:298–304.
28. Ide K, Takahashi S, Sakai K, *et al.* The dipeptide prolyl-hydroxyproline promotes cellular homeostasis and lamellipodia-driven motility via active β1-integrin in adult tendon cells. J Biol Chem 2021; 297:100819.
29. Centner C, Jerger S, Mallard A, *et al.* Supplementation of specific collagen peptides following high load resistance training upregulates gene expression in pathways involved in skeletal muscle signal transduction. Front Physiol 2022; 13:838004.
30. Iwai K, Hasegawa T, Taguchi Y, *et al.* Identification of food-derived collagen peptides in human blood after oral ingestion of gelatin hydrolysates. J Agric Food Chem 2005; 53:6531–6536.
31. Larder CE, Iskandar MM, Sabally K, Kubow S. Complementary and efficient methods for di- and tri-peptide analysis and amino acid quantification from simulated gastrointestinal digestion of collagen hydrolysate. LWT 2022; 155:112880.
32. Nogimura D, Mizushige T, Taga Y, *et al.* Prolyl-hydroxyproline, a collagen-derived dipeptide, enhances hippocampal cell proliferation, which leads to antidepressant-like effects in mice. FASEB J 2020; 34:5715–5723.

Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.